1 Words carry charge

The **service** was **terrible** *but* the **dessert** was *very* **good**

- hedons: positive charge
- dolors: negative charge
- buffer word: no charge
- operator: not, very, slightly, but

Charges come from human ratings of 16,350 words (labMT, VADER).

2 Signature of a span

19 numbers, whatever the length

words; charged words

hedons and dolors weighted by word position, degree 0 to 3

hedons and dolors under a negation, an intensifier, a diminisher and after a contrast word

sum of squared charges

3 Exact merge

σ(A) ⊕ σ(B)

σ(AB)

Counts and totals add. Position sums of B shift by the length of A.

associative · invertible · exact in integers

Holds wherever the cut falls between sentences.

4 Readouts

totals
hedons, dolors and words

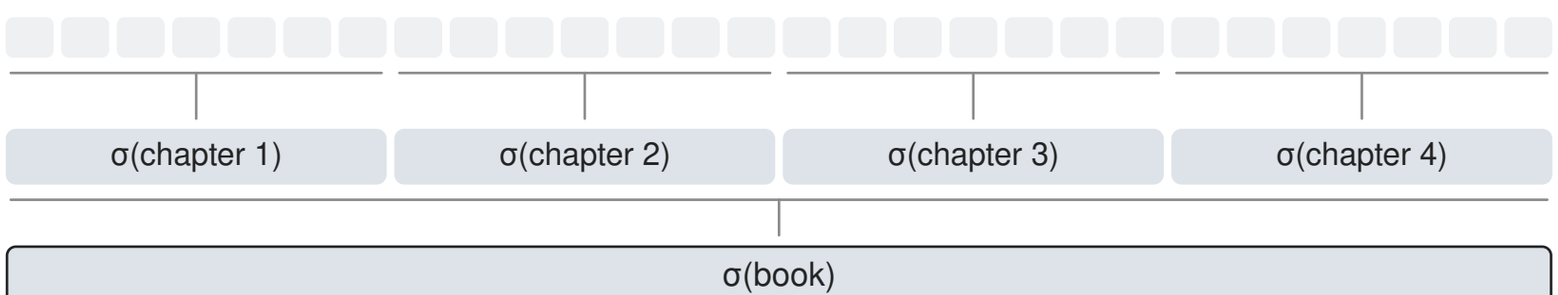
density
net charge per 1,000 words

level, with a standard error
net charge over exposure: a neutral prior, charged words and down-weighted buffer words

calibrated valence
a predicted human rating, fitted to rated text

arc
cubic of net charge over relative position

One pass, every scale



Each sentence is scanned once. Chapters and the book are merges of signatures, with no second pass over the text, and the same readouts apply at every level. A prefix table gives the totals of any run of sentences from two of its rows.